

AKSHAT SANTHANA GOPALAN

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SCHOOL / ACADEMICS

University of California, Berkeley May 2029
Data Science / Computer Science

RESEARCH PUBLICATIONS

ICLR DGM 2026: What Lies Beneath the Curve? Scaling Laws in the Presence of Exact Posteriors
***AAAI 2025:** Generative Flow Networks for Lead Optimization in Drug Design. (Association for the Advancement of Artificial Intelligence)
***IEEE MIT URTC 2025:** Deep Generative De Novo Design of High-Affinity Antibodies.
***ACS 2025:** Deep generative flow network driven multi-objective molecular design. (American Chemical Society)
***ITU AI for Good Impact Report 2025:** EquiDermAI: Inclusive Framework for Skin Disease Diagnosis (International Telecommunication Union)
ACSE CRC 2026: A Framework of RGB-Thermal Sensor Fusion with Zero-shot and Prompt-aware Deep Learning for Enhanced and Real-time Thermal Anomaly Detection in Buildings (Construction Research Congress)

*->first author

AWARDS / RECOGNITION

Regeneron Science Talent Search (STS) Scholar 2026: Top 300 Scholar nationwide (\$2,000 award).
NASA Human Exploration Rover Challenge - Top 10 International Finalist
WAICY International Finalist: AI Anomaly detection for WatchOS (LifeLine.AI).
United Nations Invited Delegate - United Nations AI for Good Global Summit 2026
International Neural Network Society - First high school student and youngest member ever inducted
Princeton University Mercer Science & Engineering Fair 2nd Place Grand Award at ISEF-Affiliated Fair
Congressional App Award - Received award from Rep. Frank Pallone Jr. for my AI Dyslexia app, BeyondWords

FUNDING / GRANTS

Air Force Research Laboratory (AFRL) \$8500 Software-Defined-Radio Grant - 2x
Currently in the process of advancing **\$400,000 STTR Grant** with Princeton University on Call for Reduced Animal Use in Research

STARTUPS

OpenScientist: RL Workflows for Automated Scientific-Discovery <https://www.theopenscientist.com/>
Segmentum AI: Spun out of Princeton; few-shot meta-segmentation models for vague boundary image; top 10% finalist at Y-Combinator <https://www.segmentum.ai/>

EXPERIENCE

SPAR AI & Sampura Research | Alignment Researcher
Sep 2026 - Present

Working with Ex-Google DeepMind Founders on benchmarks to evaluate LLM judges for scalable oversight for new UK-based based AI-Alignment NonProfit Sampura AI

Princeton Dept. of Physics | ML Researcher & Engineer
Jun 2025 - Jul 2026

Developed physics-informed tumor segmentation models and few-shot pipelines; research led to an IP filing and the commercial launch of Segmentum.

NYU Center for Data Science | ML Researcher
Jul 2025 - Dec 2026

Engineered normalizing flow systems for synthetic dataset generation and error decomposition; built GPU-intensive infrastructure to uncover deep network scaling laws.

Air Force Research Laboratory | ML & Wireless Engineer Aug 2024 - Oct 2025

Designed ML-driven optimization for LoS MIMO hardware; integrated signal modeling and simulation into a functional hardware deployment pipeline.

Stanford AIMI | Deep Learning Research Intern
Jun 2025 - Jul 2025

Built high-performance medical image segmentation systems using advanced CNNs, optimizing models for data efficiency and clinical deployment constraints.

NJIT SCIIS Lab | RL Engineer Intern
Dec 2024 - Dec 2025

Developed reinforcement-learning based crack-segmentation models, as well as RL environments for robotic navigation

EquiDerm AI | Founder
Sep 2024 - Mar 2025

Built a generative augmentation framework to eliminate racial bias in dermatological AI; recognized and published by the International Telecommunication Union and UN

IIT Madras / Cancer Institute of India | Bioinf. Intern
May 2024 - Sep 2024

Worked on creating deep-generative models for antibody design given target antigens; received SOTA results on multiple ovarian cancer antigens

Robert Bosch Center for DS and AI | ML Intern
May 2023 - Sep 2023

Created 35 deep learning models for predicting yeast-growth patterns

SKILLS / CERTIFICATIONS

Languages/Tools: Python, PyTorch, TensorFlow, JAX, Hugging Face, CUDA

Expertise: Transformers, RL (PPO, Policy Gradients), Representation Learning, Gradient-Based Optimization, Computer Vision (ViTs, Segmentation), Generative Models (Diffusion, Normalizing Flows, GFlowNets), LLM Fine-Tuning, Scientific ML

Certifications: TensorFlow Developer (Google), ML Specialization (Stanford/DeepLearning.AI), RL Specialization (UAlberta)